

BELÉN MARTÍN-URCELAY



Humboldt postdoctoral researcher at the University of Stuttgart. I study data-efficient interactive learning systems that integrate explicit and implicit human feedback, with the goal of reducing sample complexity and improving human–AI collaboration.

Research Interests: Active learning, human feedback, machine teaching, reinforcement learning, Bayesian decision making.

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RESEARCH EXPERIENCE

Humboldt Postdoctoral Researcher

University of Stuttgart, Germany

October 2026 - Present

Host: Prof. Andreas Bulling

- Develop data-efficient, adaptive human-in-the-loop learning methods that leverage implicit multimodal signals to reduce sample complexity and strengthen human–AI collaboration.

Graduate Research Assistant

Advisors: Prof. Christopher Rozell and Prof. Matthieu Bloch

September 2020 - May 2026

Georgia Institute of Technology, U.S.A.

- Expanded online machine teaching to account for learner uncertainty by simultaneously inferring the learner’s state while guiding its gradient-based learning, and proved robustness guarantees under model uncertainty.
- Improved the efficiency of knowledge transmission in human-in-the-loop learning by designing and analyzing richer forms of feedback beyond binary labels.
- Developed an interactive graph search algorithm based on posterior matching that achieves lower theoretical sample complexity bounds and implemented tree pruning, achieving up to 95% runtime reduction.

Guest Graduate Researcher

Advisors: Prof. Andreas Krause

August 2023 - December 2023

Eidgenössische Technische Hochschule (ETH), Switzerland

- Conducted research on Reinforcement Learning with Human Feedback (RLHF).
- Explored ways for Large Language Models (LLM) to interpret natural language into state-reward pairs from which to learn a reward model.

Master Thesis

Advisors: Prof. Iñaki Esnaola

February 2020 - July 2020

University of Sheffield, United Kingdom

- Developed sensor placement guidelines and experimental design methods that guarantee robustness in the information collected in a smart city environment.
- Accurately estimated missing entries employing a matrix completion technique, which led to a 7-12% error reduction for recovered entries.
- Thesis awarded distinction of Excellence by Universidad de Navarra.

Research Assistant

Advisor: Dr. Andreas Niedermeier

September 2018 - December 2018

Fraunhofer IIS, Erlangen, Germany

- Created a detector of transient signals for audio compression with Deep Neural Networks (DNN) in Python.

EDUCATION

Georgia Institute of Technology (GT)

Atlanta, U.S.A.

Ph.D. Computer Engineering

August 2022 - May 2026

- Advisors: Prof. Christopher Rozell and Prof. Matthieu Bloch.

- Topic: Efficiently teaching machine learning algorithms. Design data-efficient interactive learning systems that combine machine teaching, posterior-matching search, and rich human feedback, with robustness guarantees and provable sample-complexity improvements.

- Research Visit at ETH Zurich with Prof. Andreas Krause.

Fall 2023

M.Sc. Electrical and Computer Engineering

August 2020 - August 2022

- GPA: 4.0/4.0.

- Specialization: Signal Processing.

Universidad de Navarra (UNAV)

San Sebastián, Spain

M.Sc. Telecommunications Engineering

January 2019 - July 2020

- Grade: 9.13/10.

- Master Thesis at the University of Sheffield with Prof. Iñaki Esnaola.

Spring 2020

B.Sc. Telecommunication Systems Engineering

August 2014 - July 2018

- Grade: 8.93/10.

- Exchange semester at the University of Hong Kong. GPA: 3.94/4.

Fall 2017

TEACHING EXPERIENCE

Instructor of Record

Georgia Institute of Technology

ECE 3077 - Introduction to Probability and Statistics for ECE

January 2025 - May 2025

- Biweekly lectures followed by interactive problem-solving sessions for 60 third-year undergraduate engineering students.
- Designed a machine-learning module covering linear and logistic regression and neural networks, plus lectures, exams, and homework assignments.

Graduate Teaching Assistant

Georgia Institute of Technology

ECE2020 - Digital System Design

May 2021 - August 2021

- Conducted office hours four hours per week and independently graded assignments for 37 students.

Opportunity Research Scholars (ORS) Program

August 2020 - May 2021

- Mentored undergraduate research groups and supported fellowship and conference-proposal applications.

Undergraduate Teaching Assistant

Universidad de Navarra

Microcontrollers and Microprocessors

January 2018 - May 2018

- Facilitated lab sessions on Assembly language twice a week to 15 engineering students.

[SELECTED] INDUSTRY EXPERIENCE

ML & AI Intern

June 2026 - September 2026

California, U.S.A.

Hippo Harvest

- Developed an MCP-based interface connecting LLMs to harvest-prediction models, and designed validation, sanity-checking, and data-distribution-shift analyses to improve model reliability.

Software Developer

July 2019 - January 2020

San Sebastián, Spain

Developair

- Developed a JavaScript interface for a railway simulation tool that automated parameter adjustment.

TECHNICAL SKILLS

Languages: Spanish (native), English (bilingual, C2), Basque (medium, B1), German (basic, A2-B1)

Programming Languages:

- Python — Advanced (5 years; research projects in optimization and machine learning).
- MATLAB — Proficient (8 years; academic and research projects).
- L^AT_EX — Proficient (6 years; regularly used for typesetting academic papers and reports).
- C++ — Familiar (Visual Studio; coursework applications).

AWARDS

Scholarships

- Humboldt Research Fellowship. 2026 - 2028
- Rafael del Pino Excellency Fellowship. 2022 - 2024
- Fulbright Scholarship. 2020 - 2022
- P.E.O. International Peace Scholarship. 2020 - 2022
- Georgia Tech Electrical and Computer Engineering (ECE) Scholarship. 2020 - 2021
- International Mobility Program Connecting Talent Fellowship. 2020
- UR2PhD Technical Conference Travel Award. August 2025
- Women in Electrical and Computer Engineering (ECE) Travel grant. October 2022 and July 2025
- Travel grant by the National Science Foundation (NSF). June 2022 and June 2023

Honors

- Outstanding ECE Graduate Teaching Award, Georgia Institute of Technology. 2026
- Third poster presentation at the Georgia Tech Machine Learning Student Conference. September 2025
- Best oral presentation at the Georgia Tech Machine Learning Student Conference. September 2024
- Best Academic Record Prize by Kutxabank. 2018
- Special End of Studies Award by Universidad de Navarra. 2018
- A+ (Special mention - Top of the class) in 22 courses, 42% of the credits (ECTS). 2014 - 2020
- Summa Cum Laude
 - Graduated top of the class for master's degree. 2020
 - Graduated top of the class for bachelor's degree. 2018

PUBLICATIONS

Peer-Reviewed Journals

- *From Words to Rewards: Leveraging Natural Language For Reinforcement Learning*.
B. Martin-Urcelay, A. Krause, G. Ramponi (2026).
TMLR '26: Transactions on Machine Learning Research. January 2026.
- *Online Machine Teaching under Learner's Uncertainty: Gradient Descent Learners of a Quadratic Loss*,
B. Martin-Urcelay, C. Rozell, M. Bloch.
SIMODS '25: SIAM Journal on Mathematics of Data Science. June 2025.
- *Temperature-Dependent I/Q Imbalance Compensation in Ultra-Wideband Millimeter-Wave Multi-Gigabit Transmitters*,
A. Rezola, J. F. Sevillano, D. del Río, **B. Martin-Urcelay**, I. Gurutzeaga, I. Vélez, R. Berenguer.
IEEE Transactions on Microwave Theory and Techniques, vol. 68, no. 1, pp. 340-352, Jan. 2020.

Peer-Reviewed Conferences and Workshops

- *Representation Fragility in Reward Models: Detection and Mitigation via Sparse Autoencoders*,
S. Liu, X. Chen, **B. Martin-Urcelay**, A. Krause.
ICML '26 Workshop on Mechanistic Interpretability. July 2026.

- *MANGO: Learning Disentangled Image Transformation Manifolds with Grouped Operators*, B. Ancelin, Y. Chen, A. Saad-Falcon, P. Guan, C. Kaushik, N. Singh, **B. Martin-Urcelay**. SampTA '25 [Oral]: Sampling Theory and Applications. July 2025.
- *Enhancing Human-in-the-Loop Learning for Binary Sentiment Word Classification*, **B. Martin-Urcelay**, C. Rozell, M. Bloch. CDC '24: Conference on Decision and Control. December 2024.
- *Reinforcement Learning from Human Text Feedback: Learning a Reward Model from Human Text Input*, **B. Martin-Urcelay**, A. Krause, G. Ramponi. ICML '24 Workshop on Models of Human Feedback for AI Alignment. July 2024.

Under Review

- *Interactive Tree Search with Multiple-Choice Queries under Evaluator Noise*, **B. Martin-Urcelay**, C. Rozell, M. Bloch.

Invited Talks

- *Beyond Labels: Information-Efficient Queries to Learn from Human Feedback*, **B. Martin-Urcelay**. Keynote talk, Latinx in AI (LXAI) Workshop at ICML 2026. July 2026, Seoul, South Korea.
- *The Art of Prompt Engineering: Converting Language to Rewards in Reinforcement Learning with ChatGPT*, **B. Martin-Urcelay**. Amazon - Georgia Tech Supply Chain Research Day. February 2024, Atlanta, Georgia.
- *Human in Machine Teaching: Human and Mathematically Interpretable Query Selection*, **B. Martin-Urcelay**, C. Rozell, M. Bloch. Computer Research Association's IDEALS workshop. March 2023, Honolulu, Hawaii.

Poster Presentations

- *Teaching a Word Classifier based on Human Perception of Valence*, **B. Martin-Urcelay**, C. Rozell, M. Bloch. North American School of Information Theory (NASIT). June 2023, Philadelphia, Pennsylvania.
- *Online Machine Teaching under Learner Uncertainty*, **B. Martin-Urcelay**, C. Rozell, M. Bloch. CRA-WP Grad Cohort for Women. April 2023, San Francisco, California.
- *Online Machine Teaching with Uncertainty in Initial State*, **B. Martin-Urcelay**, C. Rozell, M. Bloch. North American School of Information Theory (NASIT). August 2022, Los Angeles, California.
- *Iterative Machine Teaching to an Unknown Learner*, **B. Martin-Urcelay**, C. Rozell, M. Bloch. CRA-WP Grad Cohort for Women. April 2022, New Orleans, Louisiana.

[SELECTED] ACADEMIC SERVICE, LEADERSHIP AND PUBLIC OUTREACH

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| Program Chair | July 2026 |
| <i>Women in Machine Learning (WiML) Symposium</i> | ICML 2026 |
| <ul style="list-style-type: none"> · Led the end-to-end peer-review process, including paper-submission management, review coordination, and communication of decisions. · Curated the invited program by selecting and coordinating speakers and panelists. | |
| Workshop Organizer | July 2025 |
| <i>ICML Workshop on Models of Human Feedback for AI Alignment</i> | ICML 2025 |
| <ul style="list-style-type: none"> · Main organizer of MoFA workshop which brings together experts in machine learning, cognitive science, behavioral psychology, and economics to explore human-AI alignment. | |

- 60 accepted papers.

Founding Graduate Chair

May 2021 - Present

Women in Electrical and Computer Engineering (WECE)

Georgia Institute of Technology

- Founded the Graduate Chair role to reach ~180 graduate women in ECE; led academic and wellness initiatives that strengthened support and belonging.
- Created volunteering opportunities to empower and promote STEM fields among local girls.

Volunteer Tutor

September 2024 - Present

Learn to be

- Provide weekly tutoring to non-English-speaking high school students from underserved communities

Graduate Student Mentor Training Course

September 2024 - December 2024

Computer Research Association (CRA)

- Learned about principles and best practices to provide effective mentorship in a research setting.

Program Coordinator

May 2021 - May 2023

Clarkston Futures Mentorship Program

Georgia Institute of Technology

- Organized STEM workshops to foster academic interest among refugee students at Clarkston High School.
- Provided after school academic support and tutoring to 20 refugee students.

Fundraiser

October 2019 - August 2023

PSE - Pour un Sourire d'Enfant

- Developed proposal strategies and forged partnerships with public and private entities to increase funding avenues.
- Successfully pitched and secured €18,995 from Fundación Pelayo, contributing to a total of €40,995 raised in 2021 for NGO projects supporting underprivileged children.